MATHEMATICAL ANALYSIS 1
HOMEWORK 14

(1) For the following ODEs, answer these questions:
(i) order (ii) linear (iii) homogeneous (iv) autonomous
(a) $y^{(5)} + (\sin x)y'' - \frac{3}{4}y = 0$
(b) $y^{(5)} + (\sin x)y'' - \frac{3}{4}x = 0$
(c) $y^{(5)} + (\sin x)y'' = \frac{3}{4}x$
(d) $y^5 + (\sin x)y'' - \frac{3}{4}x = 0$
(e) $y^{(5)} + (\sin y)y'' - \frac{3}{4}y = 0$

(2) **First-order ODEs.**
(a) $y' = x\ln(1 + x^2)$
(b) $y' = \frac{(x+2)y}{x(x+1)}$
(c) $y' = \frac{y^2}{x\ln x} - \frac{1}{x\ln x}$
(d) $y' = \sqrt[3]{2y + 3}\tan^2 x$

(3) **First-order initial-value problems.**
(a) $\begin{cases} y' = \frac{1-e^{-y}}{2x+1}, \\ y(0) = 1 \end{cases}$
(b) $\begin{cases} y' = \frac{3x}{x^2-4}|y|, \\ y(0) = -1 \end{cases}$
(c) $\begin{cases} \frac{y''}{(1+y')^{3/2}} = \frac{8x^3}{(x^4+1)^2}, \\ y(1) = 0, \quad y'(1) = 3 \end{cases}$

(4) **Second-order ODEs.**
(a) $\begin{cases} y'' + 2y' + 5y = 0, \\ y(0) = 0, \quad y'(0) = 2 \end{cases}$
(b) $\begin{cases} y'' - 4y' + 4y = 0, \\ y(0) = 1, \quad y'(0) = 0 \end{cases}$
(c) $\begin{cases} y'' - 3y' + 2y = 0, \\ y(0) = 2, \quad y'(0) = 3 \end{cases}$

HOMEWORK 14 SOLUTIONS

(1) Classification:

Part	Order	Linear	Homogeneous	Autonomous
(a)	5	Yes	Yes	No
(b)	5	Yes	No	No
(c)	5	Yes	No	No
(d)	2	No	No	No
(e)	5	No	Yes	Yes

(2) (a)

$$y = \int x \ln(1+x^2)\, dx = \frac{1}{2}(1+x^2)\ln(1+x^2) - \frac{1}{2}(1+x^2) + C.$$

(b) $y = 0$ is a one valid solution. For $y \neq 0$ we have

$$\frac{y'}{y} = \frac{x+2}{x(x+1)} = \frac{2}{x} - \frac{1}{x+1}$$

$$\ln|y| = 2\ln|x| - \ln|x+1| + C \quad \Rightarrow \quad y = C\frac{x^2}{x+1}.$$

(c) The terms with $\ln x$ mean that only $x > 0$ is valid. Observe that $y = \pm 1$ are singular (trivial) solutions (in this case $y' = 0$). Otherwise, we can separate variables:

$$\frac{y'}{y^2-1} = \frac{1}{x\ln x} \implies \int \frac{dy}{y^2-1} = \int \frac{dx}{x\ln x}$$

$$\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = \ln|\ln x| + C \implies \frac{y-1}{y+1} = C(\ln x)^2 \implies y = \frac{1+C(\ln x)^2}{1-C(\ln x)^2}.$$

(d) $y = -\frac{3}{2}$ is one constant (trivial) solution. Otherwise, we have

$$\frac{dy}{(2y+3)^{1/3}} = \tan^2 x\, dx \implies \int (2y+3)^{-1/3} dy = \int (\sec^2 x - 1) dx$$

$$\frac{3}{4}(2y+3)^{2/3} = \tan x - x + C.$$

$$y = \pm\frac{1}{2}\left[\frac{4}{3}(\tan x - x + C)\right]^{3/2} - \frac{3}{2}.$$

(3) (a) Separate: $\frac{e^y dy}{e^y - 1} = \frac{dx}{2x+1} \implies \ln|e^y - 1| = \frac{1}{2}\ln|2x+1| + C.$

$$e^y - 1 = C\sqrt{|2x+1|}.$$

Since the initial condition $y(0) = 1$ is in a neighborhood where $2x+1 > 0$, we can write:

$$e^y - 1 = C\sqrt{2x+1}.$$

Using $y(0) = 1$: $e - 1 = C(1) \implies C = e - 1$.

$$y = \ln\left(1 + (e-1)\sqrt{2x+1}\right).$$

(b) Since $y(0) = -1$, we consider the interval $x \in (-2, 2)$ where $y < 0$, so $|y| = -y$.

$$\frac{y'}{y} = -\frac{3x}{x^2-4} \implies \ln|y| = -\frac{3}{2}\ln|x^2-4| + C.$$

Using $y(0) = -1$: $\ln(1) = -\frac{3}{2}\ln(4) + C \implies C = \frac{3}{2}\ln 4 = \ln 8$.

$$|y| = \frac{8}{|x^2-4|^{3/2}} \implies y = -\frac{8}{(4-x^2)^{3/2}}.$$

(c) Let $z = y'$: $\frac{z'}{(1+z)^{3/2}} = \frac{8x^3}{(x^4+1)^2}$. Integrating both sides:

$$-2(1+z)^{-1/2} = -\frac{2}{x^4+1} + C \implies (1+z)^{-1/2} = \frac{1}{x^4+1} + C'.$$

Using $z(1) = 3$: $(1+3)^{-1/2} = \frac{1}{2} + C' \implies \frac{1}{2} = \frac{1}{2} + C' \implies C' = 0$.

$$1 + z = (x^4+1)^2 \implies z = x^8 + 2x^4.$$

Integrating z: $y = \frac{x^9}{9} + \frac{2x^5}{5} + C$. Using $y(1) = 0$: $0 = \frac{1}{9} + \frac{2}{5} + C \implies C = -\frac{23}{45}$.

$$y = \frac{x^9}{9} + \frac{2x^5}{5} - \frac{23}{45}.$$

(4) (a) $r^2 + 2r + 5 = 0 \Rightarrow r = -1 \pm 2i$. $y = e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$. $y(0) = C_1 = 0$. $y'(0) = 2C_2 = 2 \Rightarrow C_2 = 1$. $y = e^{-x} \sin 2x$.

(b) $(r-2)^2 = 0 \Rightarrow y = (C_1 + C_2 x)e^{2x}$. $y(0) = C_1 = 1$. $y' = C_2 e^{2x} + 2(C_1 + C_2 x)e^{2x}$. $y'(0) = C_2 + 2C_1 = 0 \Rightarrow C_2 = -2$. $y = (1-2x)e^{2x}$.

(c) $r^2 - 3r + 2 = 0 \Rightarrow r = 1, 2$. $y = C_1 e^x + C_2 e^{2x}$. $C_1 + C_2 = 2$ and $C_1 + 2C_2 = 3 \Rightarrow C_2 = 1, C_1 = 1$. $y = e^x + e^{2x}$.